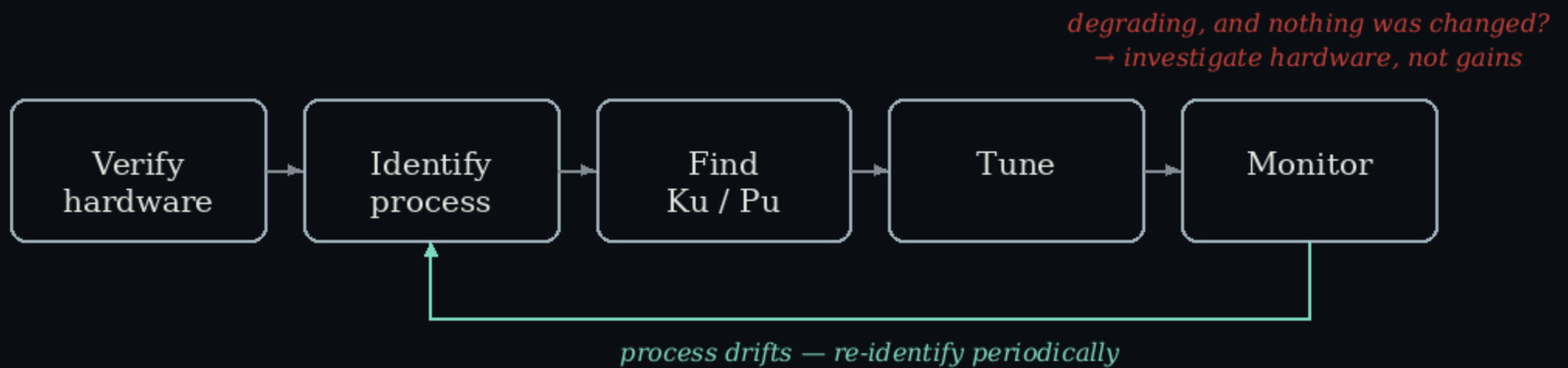




A QUICK ENGINEERING BREATH

THE LOOP NEVER REALLY CLOSES

Tuning isn't a one-time event. Commissioning is where the cycle starts.



A loop that keeps drifting back to the same problem, however carefully you retune it, is usually telling you the retuning was never going to work.

What that actually looks like — next.



A QUICK ENGINEERING BREATH

SAME SYMPTOM, TWO DIFFERENT FIXES

A trend chart can't tell you which one you have. This can.



the loop won't settle

IT'S THE MATH

Ku/Pu were wrong, or the process has genuinely drifted since it was last identified.

Fix:

re-identify, retune

IT'S THE HARDWARE

Stiction, wear, an oversized valve, a failed positioner — nothing math-shaped at all.

Fix:

repair or replace — never tune around it

The tell: was this loop fine before, and has nothing about the hardware changed since? Then it isn't the math.

Verify the hardware before you trust the tuning.